

CLINICAL PRACTICE UPDATES

AGA Clinical Practice Update on Management of Hepatitis Delta: Commentary



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DESCRIPTION: The purpose of this American Gastroenterological Association (AGA) Clinical Practice Update is to facilitate understanding and improve clinical management of patients with hepatitis delta virus infection. **METHODS:** This commentary was commissioned and approved by the AGA Institute Clinical Practice Updates Committee and the AGA Governing Board to provide timely guidance on a topic of high clinical importance to the AGA membership, and underwent internal peer review by the Clinical Practice Updates Committee and external peer review through standard procedures of *Gastroenterology*.

Keywords: Hepatitis Delta; Antivirals; Cirrhosis; Bulevirtide.

Hepatitis delta virus (HDV) infection occurs in the setting of hepatitis B virus (HBV) infection as either a coinfection or an acute superinfection in a person with underlying chronic hepatitis B (CHB).¹ The presence of concurrent HDV infection is associated with significantly increased risks of cirrhosis, hepatocellular carcinoma (HCC), and liver-related mortality compared with HBV monoinfection. Over the past decade, we have expanded our understanding of the epidemiology and management of HDV, but gaps remain. In this American Gastroenterological Association Clinical Practice Update Commentary, we provide an update on HDV epidemiology, advice for screening and diagnosis, and current and future therapies for HDV, with a focus on data in the United States.

Epidemiology of Hepatitis Delta Virus in the United States

Lack of awareness and limited testing for HDV infection have contributed to varying estimates of HDV prevalence in the United States. Furthermore, many studies have based HDV prevalence estimates on the presence of positivity to anti-HDV antibody (Ab) alone or using HDV diagnosis codes, given the lack of robust generalizable data on cohorts for which HDV RNA testing has been completed. This is due primarily to major gaps in the HDV diagnostic cascade of care, such that many people with positive anti-HDV Abs among existing large cohort studies have not subsequently completed HDV RNA testing. The presence of HDV RNA is reflective of active viremic HDV infection, and

it is estimated that approximately 50%–70% of individuals with anti-HDV positivity have detectable HDV RNA.²

Several recently published studies have provided more robust data to estimate the true prevalence of HDV in the United States. The Polaris Observatory Collaborators conducted a comprehensive literature review to estimate country-specific and population-weighted prevalence of anti-HDV and HDV RNA among *hepatitis B surface antigen* (HBsAg)-positive populations.² Areas where data are lacking were approached using a Delphi process of expert input. For the United States, the investigators used a base estimate of 1.65 million people with positive HBsAg, among those, 3.0% were estimated to be positive for anti-HDV. Among individuals positive for anti-HDV in the United States, 66.0% were estimated to have detectable HDV RNA, which equates to an estimated prevalence of 32,700 people with active HDV infection. In another recent systematic review and meta-analysis, Wong et al³ incorporated data from 329 studies that evaluated country- and region-specific HDV prevalence globally and linked these HDV prevalence estimates with data from the 2022 US Census Bureau American Community Survey. Using fixed effects meta-analyses, the prevalence of anti-HDV positivity was estimated to be 75,005, and the prevalence of viremic HDV was estimated to be 37,500–52,500 in the United States. Among non-US-born individuals with CHB, the weighted average anti-HDV prevalence was 4.20%, 45% of whom emigrated from Asia, 25% from Africa, and 14% from Europe. In high-risk populations, anti-HDV prevalence was 6.7% (95% CI, 5.2%–8.1%) among individuals with HIV coinfection, 43.8% (95% CI, 39.5%–48.0%) in people who inject drugs, and 5.1% (95% CI, 3.7%–6.5%) in men who have sex with men.³ A large US administrative claims study evaluated US adults with CHB from 2014 to 2020 using the All-Payer Claims database.⁴ Among 144,975 adults with CHB, 6719 were identified with HDV infection (4.6%) based

Abbreviations used in this paper: AASLD, American Association for the Study of Liver Diseases; Ab, antibody; aHR, adjusted hazard ratio; ALT, alanine aminotransferase; CHB, chronic hepatitis B; HBsAg, hepatitis B surface antigen; HBV, hepatitis B virus; HCC, hepatocellular carcinoma; HDV, hepatitis delta virus; NA, nucleos(t)ide analogue; peg-IFN, pegylated interferon; RR, risk ratio.

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on the *International Classification of Diseases*, ninth and tenth revisions^{5,6} diagnostic codes. This 4.6% based on the *International Classification of Diseases*, ninth and tenth revisions diagnosis codes likely reflects anti-HDV positivity, and hence when extrapolating to the US base case of 1.65 million adults with CHB, this would result in an estimated 75,900 people with positive anti-HDV and 20,240–53,130 with viremic HDV (assuming 50%–70% of anti-HDV-positive people have detectable HDV RNA). Additional published data include studies focused on US veterans,⁷ the Quest Diagnostics clinical laboratory database,⁸ and multicenter cohort studies in the United States.⁹ Taking into account these recently published data and acknowledging the limitations of interpreting the study results, given low overall HDV testing rates among individuals with HBV, we estimated the prevalence of active viremic HDV infection to be approximately 30,000–60,000 individuals in the United States.

Concurrent HDV infection among individuals with CHB is associated with significantly greater risk of disease progression to cirrhosis, hepatic decompensation, HCC, and liver-related mortality. A recent systematic review and meta-analysis inclusive of 12 studies observed that concurrent viremic HDV infection was associated with nearly double the risk of compensated cirrhosis (risk ratio [RR], 1.74; 95% CI, 1.24–2.45), a nearly 4 times higher risk of decompensated cirrhosis (RR, 3.82; 95% CI, 1.60–9.10), a 3 times higher risk of HCC (RR, 2.97; 95% CI, 1.87–4.70), and a nearly 4 times higher risk of liver-related mortality (RR, 3.78; 95% CI, 2.18–6.56) compared to those with CHB alone.¹⁰ Data from the US Department of Veterans Affairs observed similar findings.⁷ The study defined HDV infection as having positive anti-HDV or detectable RNA, and observed that those with HDV infection had a significantly greater incidence of HCC (adjusted hazard ratio [aHR], 3.61; 95% CI, 2.35–5.54), hepatic decompensation (aHR, 2.36; 95% CI, 1.50–3.70), liver-related mortality (aHR, 1.89; 95% CI, 1.25–2.86), and all-cause mortality (aHR, 1.52; 95% CI, 1.20–1.93) compared with those who did not have HDV infection.⁷ The increased risk of advanced liver disease associated with concurrent HDV infection emphasizes the importance of appropriate testing for timely diagnosis, linkage to care, and treatment.

Hepatitis Delta Virus Screening and the Diagnostic Cascade of Care

Gaps in HDV testing, which contribute to delays in HDV diagnosis, may stem in part from conflicting advice from professional societies on testing guidance. Currently, the American Association for the Study of Liver Diseases (AASLD) recommends HDV testing with anti-HDV Ab in high-risk individuals with CHB, which includes those who are HIV-positive, people who inject drugs, men who have sex with men, individuals at risk of acquiring sexually transmitted diseases, and immigrants from areas of high HDV endemicity (eg, Africa [West Africa, horn of Africa], Asia [Central and Northern Asia, Vietnam, Mongolia, Pakistan, Japan, and Taiwan], Pacific Islands [Kiribati,

Nauru], Middle East, Eastern Europe [Eastern Mediterranean regions and Turkey], South America [Amazonian basin], and Greenland.¹¹ The AASLD also suggested that HDV testing can be considered for people with CHB who have persistently low HBV DNA and elevated alanine aminotransferase (ALT). However, other professional societies, such as the European Association for the Study of the Liver and the Asian Pacific Association for the Study of the Liver recommend screening all individuals with CHB for HDV using anti-HDV Ab.^{12,13} Furthermore, one-time universal testing for HDV among adults living with CHB has been found to be cost-effective.¹⁴ We therefore suggest universal HDV screening in all patients with HBV and expect that US guidelines may be updated in the future to reflect this recommendation. Most recently, in March 2024, the World Health Organization also updated their guidance for the prevention, diagnosis, care, and treatment of individuals with CHB, and in this policy update suggested testing with anti-HDV Ab in all individuals with CHB.¹⁵ Testing for HDV should use anti-HDV Ab as the initial screening test, and individuals who are positive for anti-HDV should have follow-up testing with HDV RNA to confirm viremia, which is reflective of active HDV infection (Figure 1). When available and feasible, automated reflex testing for anti-HDV in all patients with a positive HBsAg increased anti-HDV testing rates from 8% to 93%, and decreased time to HDV testing after initial HBV result, including in a US setting.¹⁶ A double reflex test incorporates anti-HDV testing of all HBsAg-positive individuals and HDV RNA testing of all anti-HDV-positive individuals, and should be implemented to improve the diagnostic cascade of care.¹⁷ HDV antigen can be transiently detected in acute HDV infection and is not detectable in chronic infection because it forms immune complexes with anti-HDV. However, testing with HDV antigen or anti-HDV IgM is not currently advised due to their limited availability and limited sensitivity and specificity in assessing for HDV infection. Furthermore, consideration for HDV retesting should be made for those at risk of HDV superinfection, such as those individuals with ongoing risk behaviors, individuals living in areas with high HDV prevalence, and individuals with HBV who present with new liver test elevation or hepatitis B decompensation.¹⁸

Gaps in HDV testing have been reported among US populations. John et al¹⁹ evaluated 41,658 veterans with positive HBsAg and observed that only 10.7% completed HDV testing with anti-HDV or HDV RNA, which was not significantly different from a previous study 10 years earlier in the same population.²⁰ Even when evaluating individuals with high-risk features or laboratory test results that were specifically recommended for testing according to AASLD guidelines, no significant differences in HDV testing rates were observed. Similarly concerning, among the 135 individuals who were positive for anti-HDV, only 32 were tested for HDV RNA. Wong et al²¹ evaluated HDV testing patterns among 13,218 individuals with CHB across 3 US safety-net health systems. Overall, testing for anti-HDV was performed in 6.1% (4.9% among those who had risk factors specifically recommended for

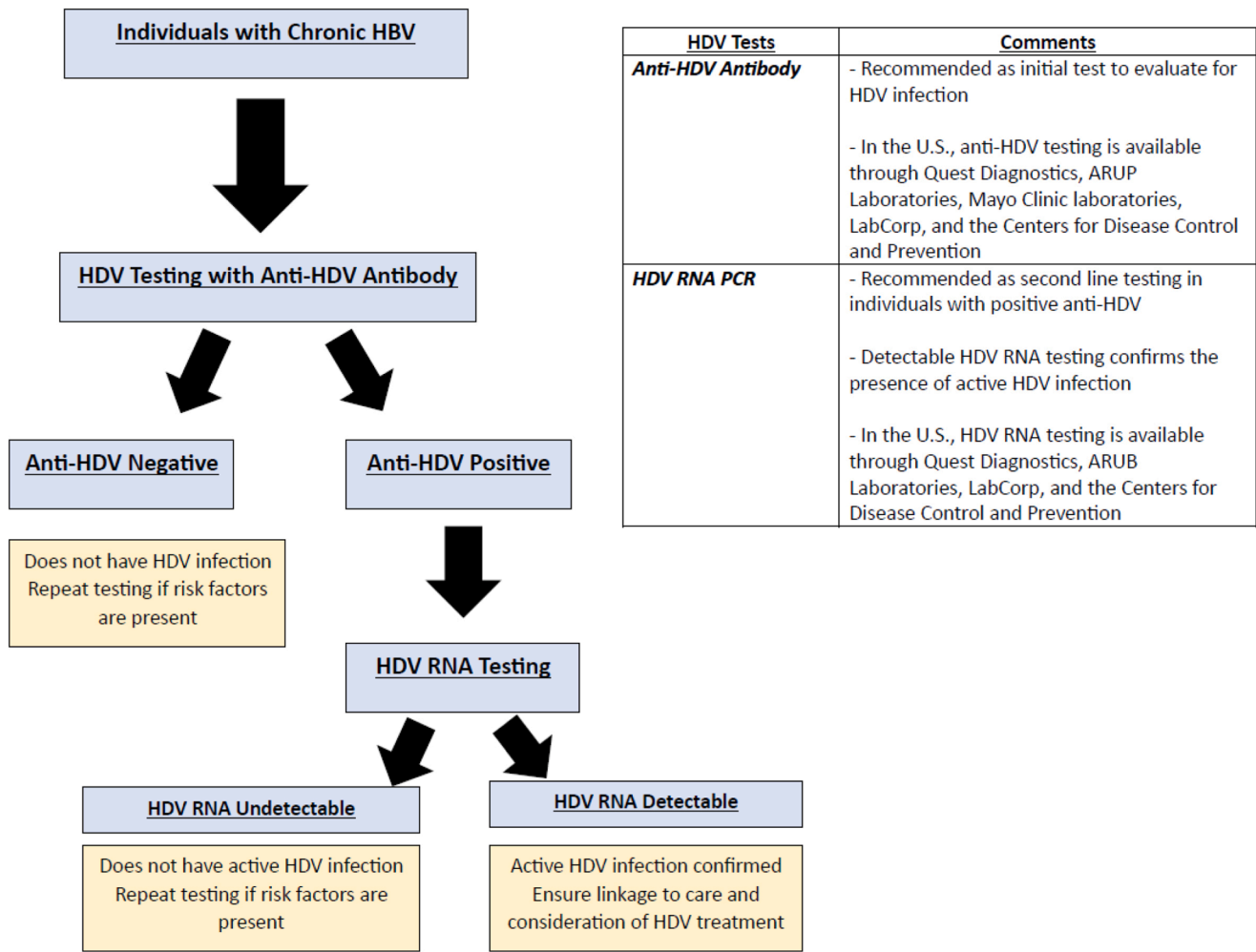


Figure 1. HDV diagnostic testing cascade.

HDV testing according to AASLD). Equally concerning was the observation that only 1.6% of the 2075 individuals positive for anti-HDV had follow-up testing with HDV RNA. Two recent studies reported on the results of implementing prospective longitudinal programs for HDV testing among individuals with CHB. Wong et al²² performed a 1-year pilot study offering universal HDV testing to all consecutive patients with CHB seen in hepatology clinics. Among 91 patients with CHB who had testing for anti-HDV Ab ordered, only 70 (76.9%) completed testing, among whom only 2 completed follow-up HDV RNA, with 1 confirmed active HDV infection. A multicenter US-based study of 1112 HBsAg-positive patients observed only 58.0% completed anti-HDV testing, and among those with positive anti-HDV, only 6 of 9 completed follow-up HDV RNA testing.⁹ These existing data raise serious concerns about gaps in HDV testing and hence delays in timely diagnosis and treatment. These gaps may reflect gaps in provider knowledge and accessibility of HDV RNA testing (which may require send out to an external laboratory and extra steps to complete during clinical practice). In the United States, HDV RNA testing is available as send-out tests via Quest Diagnostics, ARUP Laboratories, LabCorp, and the Centers for Disease

Control and Prevention (Figure 1). However, the accuracy and reliability of available HDV RNA assays are challenged by several virus-specific and assay-specific factors, which is particularly critical for HDV drug development, as well as the monitoring response to HDV treatments.²³

Monitoring of Patients With Hepatitis Delta Virus

Patients with HDV should be evaluated with imaging and noninvasive testing for the presence of cirrhosis. Vibration-controlled transient elastography has demonstrated excellent diagnostic accuracy for the detection of cirrhosis, with an area under the receiver operating characteristic curve of 0.90 compared with the aspartate aminotransferase to platelet ratio (0.83) and Fibrosis-4 (0.88). At a cutoff value of ≥ 14.0 kPa, vibration-controlled transient elastography had a sensitivity of 0.78, specificity of 0.86, negative predictive value of 0.93, and positive predictive value of 0.64 for the identification of cirrhosis in HDV. Assessment of hepatic fibrosis and prompt identification of cirrhosis through noninvasive methods (eg, Fibrosis-4 and vibration-controlled transient elastography) is a critical step in HDV management.²⁴

In line with current guidance, we advise HCC surveillance for all patients with HDV, given their increased risk of HCC.¹¹

Current and Future Treatments for Hepatitis Delta Virus

Given the particularly aggressive nature of HDV infection, it is imperative to have treatment options available for people living with HDV. Currently, the only treatment available in the United States for chronic HDV is pegylated interferon (peg-IFN)- α -2a or 2b. The sustained virologic response rate at 6 months posttreatment with peg-IFN- α ranges from 23% to 57% when treatment is administered for 48 weeks, and some studies have found that longer duration of treatment of more than 2 years can be associated with higher response rates. However, relapse after treatment discontinuation occurs in approximately 50% of patients, even after prolonged treatment, and can occur even up to 9 years after completing therapy.^{25–29} Despite incomplete response rates, there is limited but growing evidence that those who do respond have improved survival and reduced liver-related events. For example, in a small study of 12 patients who received at least 6 months of IFN therapy and were followed for 8.8 years, 4 of 5 (80%) died, compared with only 1 of 7 (14%) IFN responders.³⁰ Peg-IFN- α is contraindicated in patients with advanced liver disease or in those with major extrahepatic diseases, and is associated with significant adverse effects, including neuropsychiatric and hematologic adverse effects, which can limit its use.³¹ Per the AASLD HBV guidance statement, peg-IFN- α administered weekly for 12 months is the recommended therapy for those with elevated HDV RNA levels and elevated ALT. Furthermore, concurrent therapy with nucleos(t)ide analogues (NAs) is also recommended if HBV DNA levels are elevated (even though NAs do not have efficacy against HDV infection specifically).¹¹ The European Association for the Study of the Liver Clinical Practice Guidelines on HDV specify that NA therapy is only to be used in those patients who have an HBV DNA >2000 IU/mL and/or have cirrhosis.¹⁸ Given the high rate of relapse after peg-IFN treatment, close assessment and follow-up of HDV is necessary after treatment completion with ongoing laboratory and imaging surveillance.

Fortunately, there are newer treatment options that are currently under investigation for the treatment of HDV (Table 1).^{32–34} Bulevirtide is a sodium taurocholate co-transporting polypeptide receptor blocker, which blocks viral entry and is administered via a daily subcutaneous injection. In the phase 3 MYR-301 study, individuals with HDV were treated for 144 weeks with either 2-mg or 10-mg doses of bulevirtide, and at 48 weeks of treatment, 45% of individuals in the 2-mg arm and 48% in the 10-mg group achieved the US Food and Drug Administration-specified on-treatment response (defined by at least a 2-log decline of HDV RNA and/or ALT normalization).^{31,35} Prolonged treatment with bulevirtide can lead to a further increase in response rates.³⁶ In the MYR 204 phase 2b study,

individuals were treated with bulevirtide with and without peg-IFN- α with a primary outcome of undetectable HDV RNA at 24 weeks after the end of treatment. Up to 46% of those in the combination arm receiving bulevirtide 10 mg had undetectable HDV RNA, compared with 12% in the bulevirtide-only arm, suggesting that finite treatment with bulevirtide and peg-IFN may be considered in select patients.³⁷ Bulevirtide 2 mg has been approved for use by the European Medicines Agency since 2020, and real-world data of its use, including among patients with cirrhosis, has shown efficacy similar to clinical trials and a potential benefit on clinical outcomes.^{38,39} Although it is not US Food and Drug Administration-approved yet, and optimal treatment duration is still not defined, there is a pathway available to obtain bulevirtide 10 mg from Gilead Sciences on a case-by-case basis through an expanded access program developed to provide treatment for patients with chronic HDV and compensated liver disease with cirrhosis (Clinicaltrials.gov, Number: NCT06780579).

In addition, there are currently 2 other pharmaceutical agents that are being actively evaluated in phase 2b/phase 3 trials for the treatment of HDV at designated sites in the United States. In a phase 2 study, a monoclonal Ab targeting HBsAg called BJT-778 showed rapid achievement of a combined response in 67% of individuals treated for 24 weeks and, given these results, will be initiating a phase 2b/3 trial this year.³² In addition, targeting HBV gene expression, including messenger RNA for HBsAg, with small interfering RNAs, has been found to induce a decline in HBsAg among HBV-monoinfected patients. The phase 3 SOLSTICE study is evaluating the efficacy and safety of a combination of a monoclonal Ab, tobevibart, combined with a small interfering RNA, elebsiran, in patients with HDV. In the recently reported week-24 on-treatment data, approximately 50% of participants achieved ALT normalization, 100% achieved at least a 2-log plasma HDV RNA decline, and 41% achieved target not detected (meaning no HDV RNA amplification) in the plasma. In a small group of study participants who have been treated for 48 weeks, the latter increased to approximately 83%.³³ A global phase 3 program (ECLIPSE) evaluating the combination regimen is expected to launch in 2025. Other agents, such as lonafarnib (an orally administered inhibitor of farnesyl-transferase, a key step in the HDV viral life cycle), peg-IFN lambda (a liver-targeted peg-IFN), and nucleic acid polymers, which block HBV subviral particle assembly and inhibit HDV replication, have also been evaluated, but are not currently being investigated in phase 3 studies.³⁴

Conclusions

As we advance our understanding of HDV epidemiology in the United States, as well as improve access to HDV testing, future endeavors should focus on linkage to care and treatment. The concurrent development of novel therapeutics with increased efficacy and safety will hopefully improve disease outcomes in people living with HDV. As promising HDV treatments enter phase 3 trials and, in particular, drugs that also target a functional cure for HBV

Table 1. Current and Experimental Treatments for Hepatitis Delta Virus

Treatment	Approved in United States?	Administration	Mechanism of action	Study phase	Efficacy at wk 24/48 of treatment	SVR ^a
Peg-IFN- α -2a/2b	Yes	Weekly SC injection	Binds to IFN receptors to promote viral response	Approved	HDV RNA undetectable and ALT normal in 21%–50% at 48 wk ³²	23%–57% SVR
Bulevirtide (Gilead Sciences)	No; Expanded Access Program available (https://clinicaltrials.gov/study/NCT06780579)	Daily SC injection	Sodium taurocholate co-transporting polypeptide receptor (viral entry) inhibitor	Phase 3	45%–48% HDV RNA 2-log decline and ALT normalization at 48 wk	18% combined response at wk 48 post-treatment; 16% undetectable HDV RNA ³³
Brelovitug (BJT-778) (Bluejay Therapeutics)	No	SC injection every 1–4 wk being studied	Monoclonal antibody that targets HBsAg	Phase 2B/3	67% HDV RNA 2-log decline and ALT normalization at 24 wk; no SVR data	NA
Tobevibart and elebsiran (VIR Biotechnology)	No	2 SC injections every 2–4 wk	Monoclonal antibody that targets HBsAg and small interfering RNA	Phase 3	47% HDV RNA 2-log decline and ALT normalization at 24 wk	NA
Lonafarnib	No	Twice daily pill in combination with or without ritonavir and peg-IFN- α -2a	Farnesyltransferase (prenylation) inhibitor	Phase 3	10%–19% HDV RNA 2-log decline and ALT normalization at 48 wk	14%–26% combined response at wk 24 post-treatment; 6%–23% undetectable HDV RNA ³⁴
Peg-IFN- λ	No	Weekly SC injection	Binds to IFN receptors to promote viral response	Phase 3, discontinued	NA	NA
Nucleic acid polymer (REP2139)	No	SC injection weekly with or without peg-IFN	Blocks HBV subviral particle assembly	Phase 2/ compassionate use in Europe	73% HDV RNA 2-log decline, 61% HDV RNA negative	NA

SC, subcutaneous; SVR, sustained virologic response.

^aDefined as undetectable HDV RNA off treatment.

are developed, the future will hopefully bring curative treatments for HDV that can decrease disease burden and optimize outcomes for people living with HDV.

Video

To view an accompanying video interview with the authors, please visit the online version of the article at <https://doi.org/10.1053/j.gastro.2025.07.037>.

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Author Contributions

All co-authors developed the concept and design for this article. Tatyana Kushner and Robert J. Wong drafted the initial manuscript. Stanley M. Cohen and Joseph Ahn provided critical revisions of the manuscript. All co-authors reviewed the final version of the manuscript.

Conflicts of interest

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